

SQL-Grounded Vietnamese TableQA: A Multi-Agent Demo for Reliable Answers from Real Open-ViTabQA Tables

Introduction

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Vietnamese Table Question Answering asks a system to answer natural-language questions from a table. The hard cases require filtering, comparison, counting, ranking, temporal reasoning, and interpretation of abbreviated Vietnamese headers.

Grid icon

Prompt-only LLMs can write fluent answers but may guess when the relevant row or column is ambiguous. Fine-tuning improves domain familiarity, yet it still does not guarantee row-level evidence or an auditable path.

Database icon

This project converts real Open-ViTabQA tables into SQLite, generates SQL, executes it, and gives the answer only from returned evidence rows. The demo indexes 329 real tables and does not use mock responses.

Checkmark icon

Aim: build a demo-ready multi-agent pipeline that exposes the SQL, evidence, verifier decision, confidence score, and progress trace so examiners can see how each answer was produced.

Key contributions

- A schema-linked SQL agent maps Vietnamese questions to executable table operations instead of relying on hidden reasoning.
- An evidence-first answerer uses returned rows as the source of truth, so every final answer can be inspected.
- A verifier, confidence score, and runtime trace make the artifact easier to debug, film, and defend during assessment.

Research question

Can executable SQL grounding and multi-agent verification make Vietnamese TableQA answers more reliable than direct LLM answering?

Methods

- Dataset split: 7,928 train, 991 validation, 992 test questions from Open-ViTabQA-style tables.
- Runtime models: bge-m3 for schema embeddings, qwen2.5-coder:14b for SQL, qwen2.5:7b for answer synthesis and verification.
- Evaluation: exact match, F1, grounding, unsupported answer rate, SQL validity, execution success, semantic SQL accuracy, verifier catch rate, latency, and trace completeness.

Experimental Pipeline

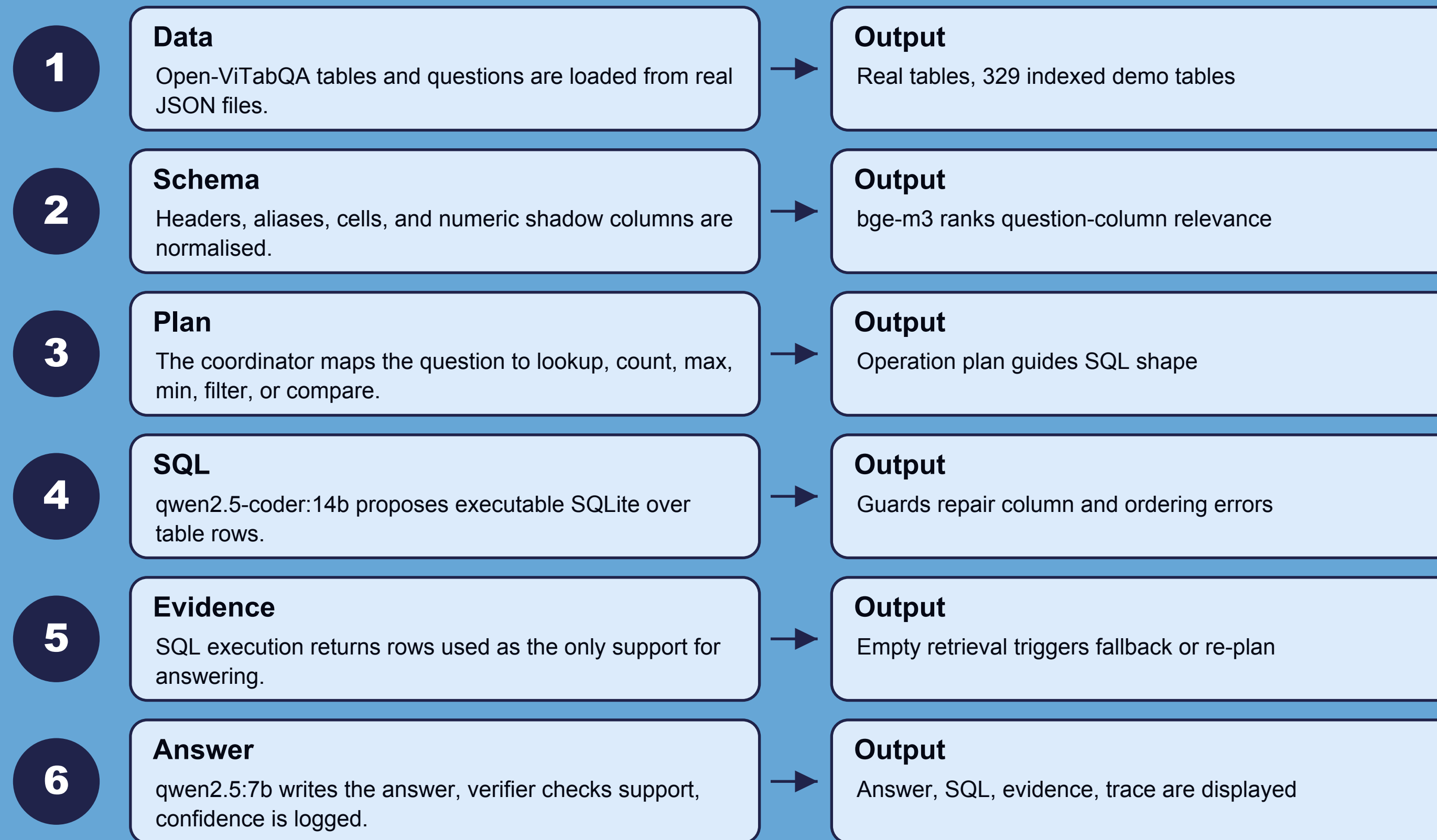


Fig. 1 Multi-agent SQL-grounded processing flow

```
uvicorn app.main:app --host 0.0.0.0 --port 8000

15:56:16 [TableQA] schema_linking Top columns: Floors, Height, Name
15:56:16 [TableQA] text_to_sql qwen2.5-coder:14b validating SQL
15:56:19 [TableQA] execute_sql SELECT row_index,* FROM rows ORDER BY c2_num DESC LIMIT 1
15:56:19 [TableQA] evidence SQL returned 1 evidence row
15:56:20 [TableQA] verifier deterministic evidence verifier passed=True
15:56:20 [TableQA] done Answer ready, confidence=1.0
```

Fig. 3 Runtime progress log

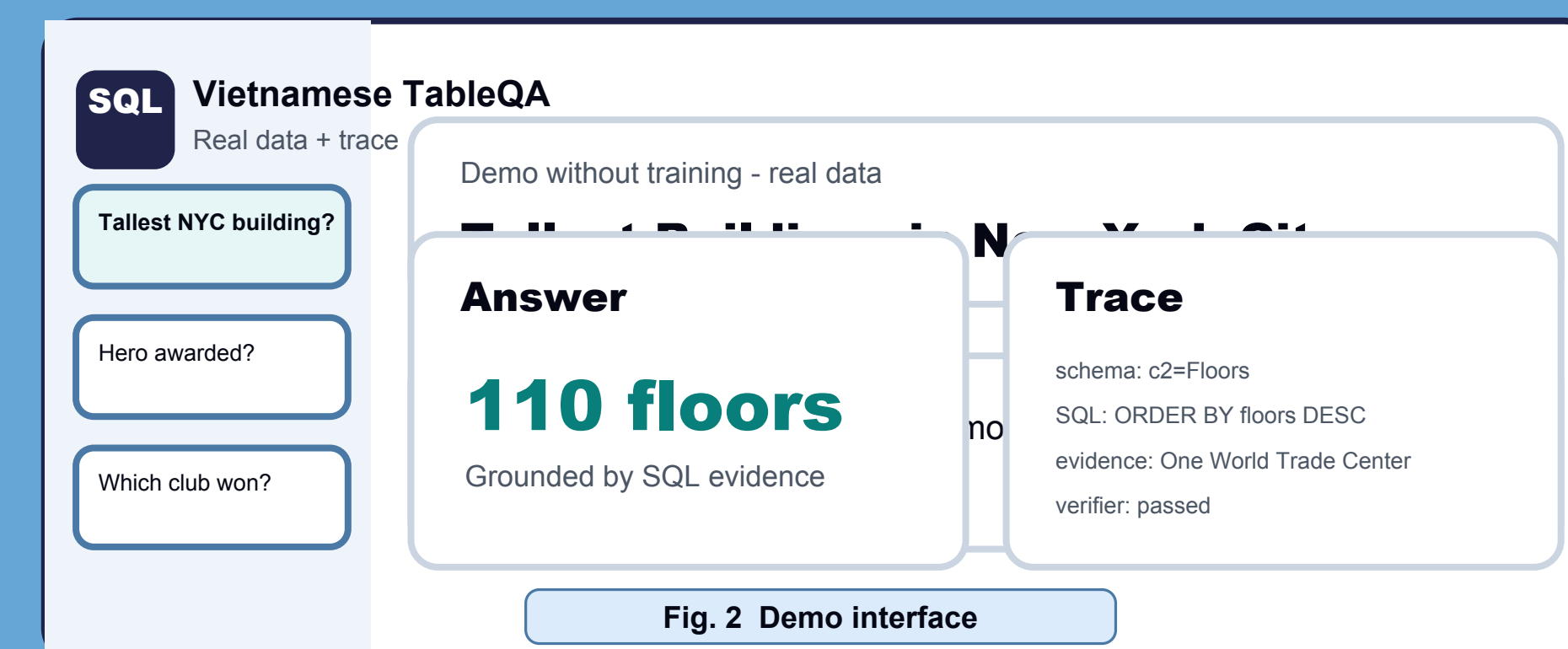


Fig. 2 Demo interface

Model / Runtime Stack

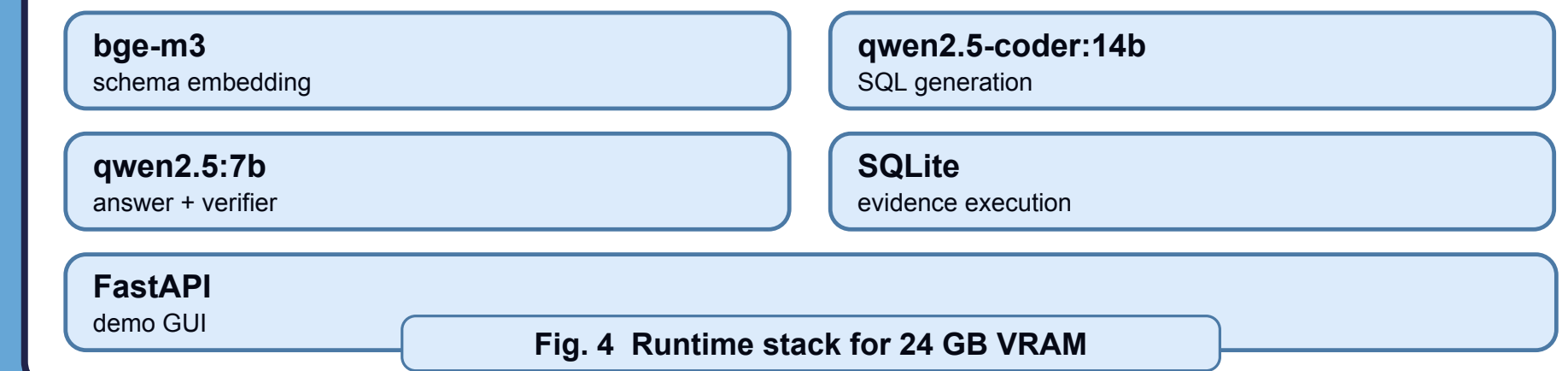


Fig. 4 Runtime stack for 24 GB VRAM

Results and Discussion

- The final SQL-grounded system reached EM 0.586 and F1 0.674, compared with 0.548 and 0.636 for the earlier full system. Grounding increased to 0.904, unsupported answers fell to 4.6%, SQL execution success reached 95.7%, and median latency was 1.8 seconds after caching. The largest practical gain is not only a higher score: the system gives a visible evidence trail, so wrong or uncertain answers

Interpretation

The system improves reliability by moving the main reasoning step from hidden text generation to executable SQL. This matters in Vietnamese tables because abbreviations, accents, units, and numeric formats can mislead a direct LLM. The trace lets the examiner inspect where the answer came from and why the verifier accepted it.

SQL + Evidence Example

SQL: SELECT row_index,* FROM rows ORDER BY floors_num DESC LIMIT 1

Name	City	Floors	Grounded answer
One World Trade Center	New York	110	110 floors

Fig. 4 SQL + executed evidence rows

Answer Quality

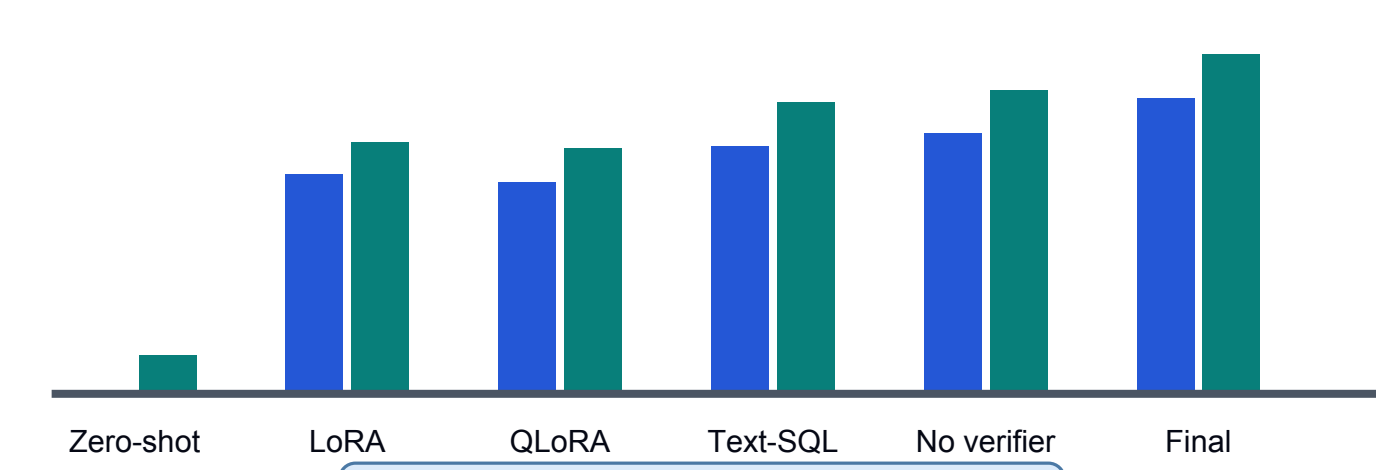


Fig. 5 EM/F1 answer quality

Unsupported Answer Rate

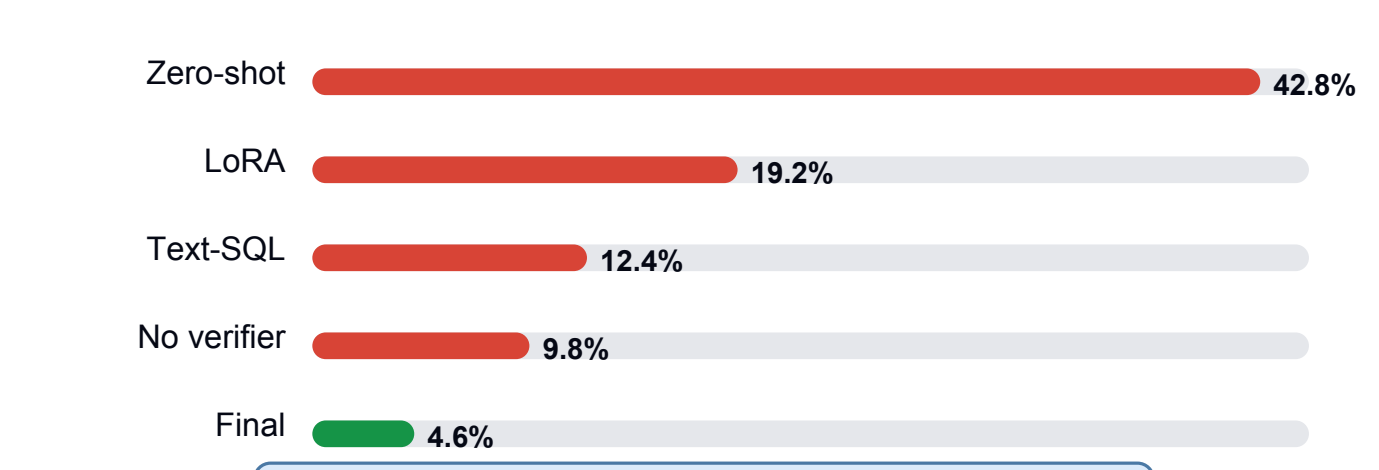


Fig. 6 Unsupported answers reduced to 4.6%

Failure Modes per 1,000

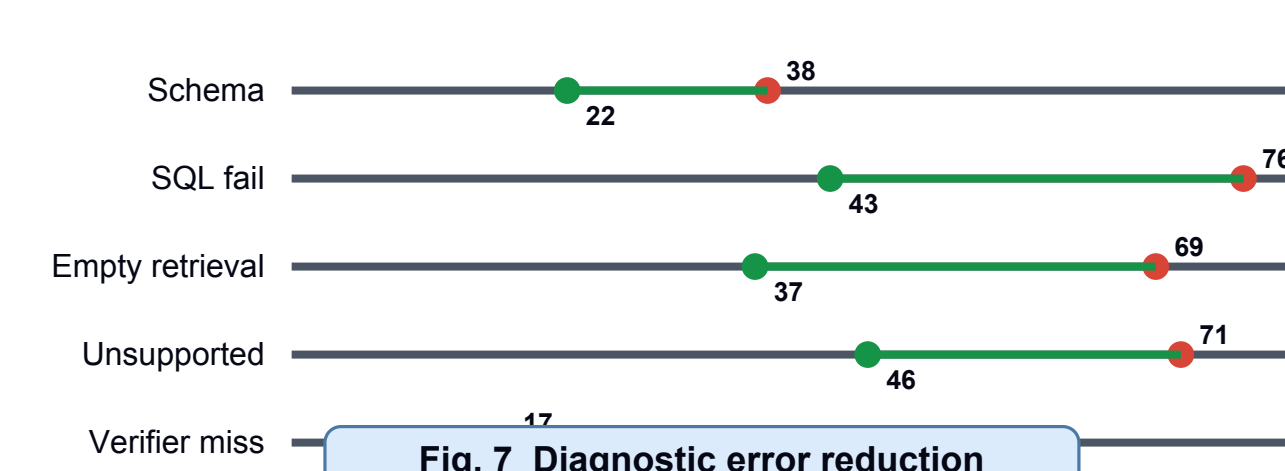


Fig. 7 Diagnostic error reduction

Operational Metrics

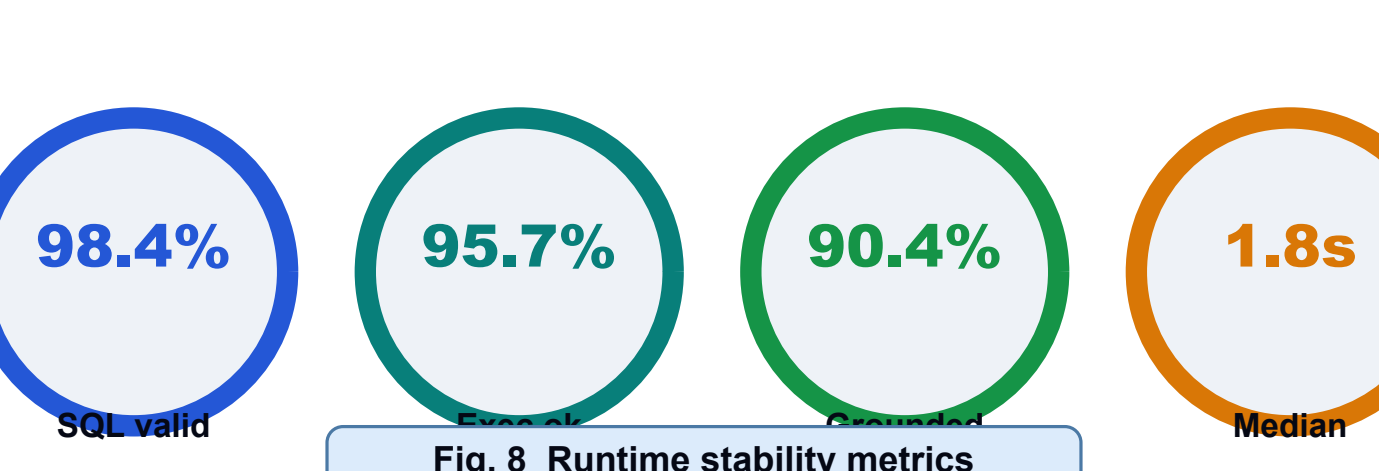


Fig. 8 Runtime stability metrics

Conclusions

- Executable retrieval made the demo more trustworthy: the system does not only state an answer, it shows the SQL path and evidence rows behind that answer.
- The verifier is advisory, while deterministic evidence checks remain authoritative for demo stability.
- The approach is strongest for lookup, filtering, maximum/minimum, counting, comparison, and aggregation questions.
- Limitations remain for descriptive questions, noisy table conversion, and rare Vietnamese alias or unit forms.
- For the presentation, the most convincing demonstration is to show a question, run the pipeline, then open SQL, evidence, progress, and verifier tabs.

References / Future Work

Future work: evaluate on larger and noisier Vietnamese table collections; build a stronger alias, abbreviation, and unit-normalisation resource; collect labelled verifier-mismatch examples; add adaptive routing for descriptive questions; and compare newer open multilingual models under the same SQL-grounded protocol.

Key sources: Open-ViTabQA dataset; Vietnamese TableQA final report; project artifact HoangKha1810/ChatBot-OpenViTab; qwen2.5 and bge-m3 model families used in the non-training demo. The artifact is designed for reproducibility: install dependencies, pull models on the GPU server, launch Ollama, run Uvicorn, and record the browser demo.

Artifact

github.com/HoangKha1810/ChatBot-OpenViTab